Alexander Turco

Doctoral Researcher PhD(c), Medical Biophysics · University of Toronto

Computational Cancer Genomics · Non-B DNA · Genomic Variation · Tandem Repeats · Machine Learning

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Education

University of Toronto (Princess Margaret Cancer Centre) — PhD in Medical Biophysics Sep 2024–Present
Thesis Supervisor: Dr. Sushant Kumar

- PhD Thesis: Repetitive Genomic Elements as Multi-Scale Vulnerabilities in Cancer: Examining DNA Structure, Sequence, and Protein-Level Repetition

McMaster University — Honours BSc in Biology, Research Specialization Sep 2019–Dec 2023
Thesis Supervisors: Dr. G. Brian Golding, Dr. Rosa da Silva

- Golding Lab Thesis: Estimating Evolutionary Parameters for Protein Low Complexity Regions using an Approximate Bayesian Computation
- da Silva Lab Thesis: Cells at War: The Playfulness of Game-Based Learning

Research Appointments & Experience

Doctoral Researcher — Kumar Lab, Computational Cancer Genomics Sep 2024–Present
University of Toronto / Princess Margaret Cancer Centre, UHN *Toronto, ON*

- Characterize non-B DNA motifs across 130 haplotype-resolved human genome assemblies, examining their stability, genomic localization, and relationships with structural variation.
- Develop scalable computational workflows to study genomic repeats, regulatory elements, and population-scale human genomic variation.
- Investigate tandem-repeat expansions across more than 16,000 matched tumour-normal whole-genome sequencing samples spanning 38 cancer types.
- Develop machine learning approaches for detecting G-quadruplex DNA structures using short-read and nanopore sequencing data.

Research Technician I — Kumar Lab, Computational Cancer Genomics Jan 2024–Aug 2024
Princess Margaret Cancer Research Centre, University Health Network *Toronto, ON*

- Investigated non-B DNA structures using Oxford Nanopore sequencing and population-scale genomic analyses.
- Developed a pipeline for nanopore basecalling, read alignment, and extraction of per-base translocation times from long-read sequencing data.
- Built a machine learning model to predict non-B DNA structures from nanopore sequencing-derived signals.

Research Student — Kumar Lab, Computational Cancer Genomics May 2023–Aug 2023
Princess Margaret Cancer Research Centre, University Health Network *Toronto, ON*

- Investigated sex-associated gene expression differences across 12 human cancers using TCGA RNA-seq data.
- Developed a computational workflow for differential expression analysis and identification of candidate synthetic lethal interactions.
- Integrated gene expression and interaction data to examine potential sex-specific cancer vulnerabilities.

Research Student — Golding Lab, Bioinformatics and Molecular Evolution May 2022–Apr 2023
Department of Biology, McMaster University *Hamilton, ON*

- Analyzed 16S rRNA amplicon sequencing data to characterize microbial communities in freshwater algal blooms across Ontario.
- Applied R, Python, and Unix tools for sequence analysis, computational biology, and data visualization.
- Developed a `C++` simulation of protein low-complexity region evolution for parameter inference using Approximate Bayesian Computation.

Research Student — da Silva Lab, Pedagogy and Science Education May 2022–Apr 2023
Department of Biology, McMaster University *Hamilton, ON*

- Investigated game-based learning through the development and evaluation of the educational biology game *Cells at War*.
- Designed student feedback surveys and analyzed responses to assess engagement, motivation, and perceptions of game-based learning.
- Collaborated with an international interdisciplinary team on educational game development and communicated findings through research presentations.

Publications & Scholarly Works

Turco, A., Boev, N. & Kumar, S. (2026). *Diversity and genomic organization of non-B DNA motifs in haplotype-resolved human genome assemblies*. bioRxiv. doi:10.64898/2026.03.05.709836.

Turco, A., Flaim, B., Ponzi, G., Ayekun, N., Amore, J-P & da Silva, R. (2023). *A STEAM game-based learning framework: Maximizing integrated and immersive learning in the classroom*. Western Conference on Science Education (conference abstract). Publication link.

Awards & Honours

Marathon of Hope Health Informatics & Data Science Award Sep 2026
Marathon of Hope Cancer Centres Network • $50,000

Peterborough K.M. HUNTER Charitable Foundation Graduate Award Jul 2026
Temerty Faculty of Medicine, University of Toronto • $23,000

SGS Conference Grant May 2026
School of Graduate Studies, University of Toronto • $350

MBP Excellence OSOTF PMH Award Aug 2025
Ontario Cancer Institute & Princess Margaret Cancer Centre • $5,000

Canada Graduate Scholarship Masters (CGS-M) May 2025
Canadian Institutes of Health Research • $27,000

Best Poster Award Apr 2025
Toronto DNA Replication and Repair Symposium, University of Toronto • $100

Oral Presentation Awards in Computational Biology & Science Education Apr 2023
Biology Undergraduate Symposium, McMaster University

3rd Place Abstract Award Oct 2022
MacWater Challenges in Water Monitoring Conference, McMaster University

Research stipend for the creation of Cells at War: A Biological Video Game Sep 2021–Dec 2023
Co-operative Education and Work-Integrated Learning Canada (CEWIL) • $6,000

Invited Seminars

Non-B DNA structures, evolution, and genomic instability Sep 2025
Invited speaker • Buffalo Evolutionary and Anthropological Genomics Lab (BEAGL) • University at Buffalo, NY

Diversity and Genomic Organization of Non-B DNA structures across 130 haplotype-resolved human assemblies Aug 2025
Invited speaker • Human Genomic Structural Variation Consortium (HGSVC) • Virtual

Conference Contributions

Non-B DNA structures in haplotype-resolved human genome assemblies Oct 2026
Poster • American Society of Human Genetics (ASHG) • Montréal, QC

Non-B DNA structures in haplotype-resolved human genome assemblies Oct 2026
Oral presentation • American Society of Human Genetics (ASHG) • Montréal, QC

Tandem Repeat Expansions Drive Genomic Instability and Clinical Outcomes Across Cancers May 2026
Poster • Canadian Bioinformatics Hub Conference (CBHC, ISCB) • Toronto, ON

Decoding the Cancer Repeatome: Nanopore sequencing based discovery of Non-B DNA motifs May 2025
Poster • Medical Biophysics Graduate Student Symposium • Toronto, ON

T2T Assemblies Reveal Non-B DNA's Influence on Genome Integrity Apr 2025
Poster, Best Poster Award • Toronto DNA Replication and Repair Symposium • Toronto, ON

Cells at War: Maximizing integrated and immersive learning in the classroom Jul 2023
Oral Presentation • Western Conference on Science Education • London, ON

Estimating Evolutionary Parameters for Protein Low Complexity Regions using an ABC Apr 2023
Oral presentation • Biology Undergraduate Symposium, McMaster University • Hamilton, ON

16S rRNA analysis of harmful algal blooms in Ontario Oct 2022
Poster • MacWater Challenges in Water Monitoring Conference • Hamilton, ON
Additional meeting attended: Telomere-to-Telomere Face-to-Face Conference, University of California Santa Cruz (Sep 2024).

Teaching, Mentorship & Community

Networking Team Member — Life Sciences Career Development Syndicate 2026–Present
Recruit life sciences professionals and coordinate large-scale career development and networking events attended by hundreds of students, connecting trainees with diverse career pathways across the life sciences sector.

Undergraduate Research Mentor — BCB330Y / BCB430Y, University of Toronto 2025–Present
Mentored two undergraduate students (one annually) on independent research projects in bioinformatics and computational biology, providing guidance on research design, computational analysis, and scientific communication.

President — Medical Biophysics Career Development Association, University of Toronto 2025–2026
Led departmental career development initiatives, organizing professional networking events, recruiting alumni from diverse sectors, and collaborating with student organizations to secure funding and support student pitch competitions.

Technical Skills

Programming & Computing: Python, R, C++, Bash, Unix/Linux, Git/GitHub, SLURM / high-performance computing, LaTeX.

Genomics & Bioinformatics: SPAdes, Kraken2, BLAST, fastp, SAMtools/BAMtools, GSEA, DESeq2, BWA, minimap2, Dorado, Remora, Verkko, hifiasm, IGV.

Machine Learning & Data Analysis: PyTorch, Scanpy, AnnData, PCA, Bioconductor, tidyverse, ggplot2, dplyr.